

Exosome Circuitry During (De)(Re)Myelination of the Central Nervous System.

PMID: 32612996 | Indexed in PubMed

Reciprocal neuron-glia cell communication is fundamental for the proper function of the nervous system. Oligodendrocytes are the myelinating cells of the central nervous system (CNS) that insulate and provide trophic support to neurons. This effective interaction is crucial not only for myelination but also for long-term axonal survival and neural connectivity. In recent years, exosomes have been portrayed as key players in intercellular interaction in the context of the healthy and diseased CNS. They act as communicating vehicles, true attachés operating between neurons and glial cells. Despite the complex exosome circuitry within the nervous system, experimental evidence supports the role of exosomes in modulating myelination. Oligodendrocytes secrete exosomes in response to neuronal signals in an electric activity-dependent manner. These released exosomes are then internalized by neurons, contributing to their integrity and activity. In turn, neurons secrete exosomes to control the communication between them and with myelinating cells in order to regulate synaptic function in neuronal development, myelin maintenance, and neuroregeneration. In this review, we provide a critical view of the current understanding on how exosomes, either from CNS-resident cells or from the periphery, contribute to the formation and maintenance of myelin and, additionally, on how the differential content of exosomes in normal and pathological conditions foresees the use of these nanovesicles as putative diagnostic and/or therapeutical agents in white matter degeneration-associated diseases.

Inwardly rectifying potassium channels (Kir) in central nervous system glia: a special role for Kir4.1 in glial functions.

PMID: 16563220 | Indexed in PubMed

Glia in the central nervous system (CNS) express diverse inward rectifying potassium channels (Kir). The major function of Kir is in establishing the high potassium (K⁺) selectivity of the glial cell membrane and strongly negative resting membrane potential (RMP), which are characteristic physiological properties of glia. The classical property of Kir is that K⁺ flows inwards when the RMP is negative to the equilibrium potential for K⁺ (E(K)), but at more positive potentials outward currents are inhibited. This provides the driving force for glial uptake of K⁺ released during neuronal activity, by the processes of "K⁺ spatial buffering" and "K⁺ siphoning", considered a key function of astrocytes, the main glial cell type in the CNS. Glia express multiple Kir channel subtypes, which are likely to have distinct functional roles related to their differences in conductance, and sensitivity to intracellular and extracellular factors, including pH, ATP, G-proteins, neurotransmitters and hormones. A feature of CNS glia is their specific expression of the Kir4.1 subtype, which is a major K⁺ conductance in glial cell membranes and has a key role in setting the glial RMP. It is proposed that Kir4.1 have a primary function in K⁺ regulation, both as homomeric channels and as heteromeric channels by co-assembly with Kir5.1 and probably Kir2.0 subtypes. Significantly, Kir4.1 are also expressed by oligodendrocytes, the myelin-forming cells of the CNS, and the genetic ablation of Kir4.1 results in severe hypomyelination. Hence, Kir, and in particular Kir4.1, are key regulators of glial functions, which in turn determine neuronal excitability and axonal conduction.

Evolvability of the actin cytoskeleton in oligodendrocytes during central nervous system

development and aging.

PMID: 30302529 | Indexed in PubMed

The organization of actin filaments into a wide range of subcellular structures is a defining feature of cell shape and dynamics, important for tissue development and homeostasis. Nervous system function requires morphological and functional plasticity of neurons and glial cells, which is largely determined by the dynamic reorganization of the actin cytoskeleton in response to intrinsic and extracellular signals. Oligodendrocytes are specialized glia that extend multiple actin-based protrusions to form the multilayered myelin membrane that spirally wraps around axons, increasing conduction speed and promoting long-term axonal integrity. Myelination is a remarkable biological paradigm in development, and maintenance of myelin is essential for a healthy adult nervous system. In this review, we discuss how structure and dynamics of the actin cytoskeleton is a defining feature of myelinating oligodendrocytes' biology and function. We also review "old and new" concepts to reflect on the potential role of the cytoskeleton in balancing life and death of myelin membranes and oligodendrocytes in the aging central nervous system.

Glutamate-mediated glial injury: mechanisms and clinical importance.

PMID: 16206168 | Indexed in PubMed

Primary and/or secondary glial cell death can cause and/or aggravate human diseases of the central nervous system (CNS). Like neurons, glial cells are vulnerable to glutamate insults. Astrocytes, microglia, and oligodendrocytes express a wide variety of glutamate receptors and transporters that mediate many of the deleterious effects of glutamate. Astrocytes are responsible for most glutamate uptake in synaptic and nonsynaptic areas and consequently, are the major regulators of glutamate homeostasis. Microglia in turn may secrete cytokines, which can impair glutamate uptake and reduce the expression of glutamate transporters. Finally, oligodendrocytes, the myelinating cells of the CNS, are very sensitive to excessive glutamate signaling, which can lead to the apoptosis or necrosis of these cells. This review aims at summarizing the mechanisms leading to glial cell death as a consequence of alterations in glutamate signaling, and their clinical relevance. A thorough understanding of these events will undoubtedly lead to better therapeutic strategies to treat CNS diseases affecting glia and in particular, those that involve damage to white matter tracts.

The Shh receptor Boc is important for myelin formation and repair.

PMID: 31048318 | Indexed in PubMed

Myelination leads to the formation of myelin sheaths surrounding neuronal axons and is crucial for function, plasticity and repair of the central nervous system (CNS). It relies on the interaction of the axons and the oligodendrocytes: the glial cells producing CNS myelin. Here, we have investigated the role of a crucial component of the Sonic hedgehog (Shh) signalling pathway, the co-receptor Boc, in developmental and repairing myelination. During development, Boc mutant mice display a transient decrease in oligodendroglial cell density together with delayed myelination. Despite recovery of oligodendroglial cells at later stages, adult mutants still exhibit a lower production of myelin basic protein correlated with a significant decrease in the calibre of callosal axons and a reduced amount of the

neurofilament NF-M. During myelin repair, the altered OPC differentiation observed in the mutant is reminiscent of the phenotype observed after blockade of Shh signalling. In addition, Boc mutant microglia/macrophages unexpectedly exhibit the apparent inability to transition from a highly to a faintly ramified morphology in vivo. Altogether, these results identify Boc as an important component of myelin formation and repair.

New insights into signaling during myelination in zebrafish.

PMID: 22074600 | Indexed in PubMed

Myelin is a vertebrate adaptation that allows for the rapid propagation of action potentials along axons. Specialized glial cells-oligodendrocytes in the central nervous system (CNS) and Schwann cells in the peripheral nervous system (PNS)-form myelin by repeatedly wrapping axon segments. Debilitating diseases result from the disruption of myelin, including multiple sclerosis and Charcot-Marie-Tooth peripheral neuropathies. The process of myelination involves extensive communication between glial cells and the associated neurons. The past few years have seen important progress in understanding the molecular basis of the signals that coordinate the development of these fascinating cells. This review highlights recent advances in myelination deriving from studies in the zebrafish model system, with a primary focus on the PNS. While Neuregulin1-ErbB signaling has long been known to play important roles in peripheral myelin development, work in zebrafish has elucidated its roles in Schwann cell migration and radial sorting of axons in vivo. Forward genetic screens in zebrafish have also uncovered new genes required for development of myelinated axons, including *gpr126*, which encodes a G-protein coupled receptor required for Schwann cells to progress beyond the promyelinating stage. In addition, work in zebrafish uncovered new roles for Schwann cells themselves, including in regulating the boundary between the PNS and CNS and positioning a nerve after its initial outgrowth.

Exploring Ca²⁺ Dynamics in Myelinating Oligodendrocytes through rAAV-Mediated jGCaMP8s Expression in Developing Spinal Cord Organ Cultures.

PMID: 38744490 | Indexed in PubMed

Oligodendrocytes, the myelin-producing glial cells of the central nervous system (CNS), crucially contribute to myelination and circuit function. An increasing amount of evidence suggests that intracellular calcium (Ca²⁺) dynamics in oligodendrocytes mediates activity-dependent and activity-independent myelination. Unraveling how myelinating oligodendrocytes orchestrate and integrate Ca²⁺ signals, particularly in relation to axonal firing, is crucial for gaining insights into their role in the CNS development and function, both in health and disease. In this framework, we used the recombinant adeno-associated virus/Olig001 capsid variant to express the genetically encoded Ca²⁺ indicator jGCaMP8s, under the control of the myelin basic protein promoter. In our study, this tool exhibits excellent tropism and selectivity for myelinating and mature oligodendrocytes, and it allows monitoring Ca²⁺ activity in myelin-forming cells, both in isolated primary cultures and organotypic spinal cord explants. By live imaging of myelin Ca²⁺ events in oligodendrocytes within organ cultures, we observed a rapid decline in the amplitude and duration of Ca²⁺ events across different in vitro developmental stages. Active myelin sheath remodeling and growth are modulated at the level of myelin-axon interface through Ca²⁺ signaling, and, during early myelination in organ cultures, this phase

is finely tuned by the firing of axon action potentials. In the later stages of myelination, Ca²⁺ events in mature oligodendrocytes no longer display such a modulation, underscoring the involvement of complex Ca²⁺ signaling in CNS myelination.

CNS myelin sheath is stochastically built by homotypic fusion of myelin membranes within the bounds of an oligodendrocyte process.

PMID: 25682762 | Indexed in PubMed

Myelin - the multilayer membrane that envelops axons - is a facilitator of rapid nerve conduction. Oligodendrocytes form CNS myelin; the prevailing hypothesis being that they do it by extending a process that circumnavigates the axon. It is pertinent to ask how myelin is built because oligodendrocyte plasma membrane and myelin are compositionally different. To this end, we examined oligodendrocyte cultures and embryonic avian optic nerves by electron microscopy, immuno-electron microscopy and three-dimensional electron tomography. The results support three novel concepts. Myelin membranes are synthesized as tubules and packaged into "myelinophore organelles" in the oligodendrocyte perikaryon. Myelin membranes are matured in and transported by myelinophore organelles within an oligodendrocyte process. The myelin sheath is generated by myelin membrane fusion inside an oligodendrocyte process. These findings abrogate the dogma of myelin resulting from a wrapping motion of an oligodendrocyte process and open up new avenues in the quest for understanding myelination in health and disease.

Developmental and Repairing Production of Myelin: The Role of Hedgehog Signaling.

PMID: 30237763 | Indexed in PubMed

Since the discovery of its role as a morphogen directing ventral patterning of the spinal cord, the secreted protein Sonic Hedgehog (Shh) has been implicated in a wide array of events contributing to the development, maintenance and repair of the central nervous system (CNS). One of these events is the generation of oligodendrocytes, the glial cells of the CNS responsible for axon myelination. In embryo, the first oligodendroglial cells arise from the ventral ventricular zone in the developing brain and spinal cord where Shh induces the basic helix-loop-helix transcription factors Olig1 and Olig2 both necessary and sufficient for oligodendrocyte production. Later on, Shh signaling participates in the production of oligodendroglial cells in the dorsal ventricular-subventricular zone in the postnatal forebrain. Finally, the modulation of Hedgehog signaling activity promotes the repair of demyelinated lesions. This mini-review article focuses on the Shh-dependent molecular mechanisms involved in the spatial and temporal control of oligodendrocyte lineage appearance. The apparent intricacy of the roles of two essential components of Shh signaling, Smoothened and Gli1, in the postnatal production of myelin and its regeneration following a demyelinating event is also highlighted. A deeper understanding of the implication of each of the components that regulate oligodendrogenesis and myelination should beneficially influence the therapeutic strategies in the field of myelin diseases.

Oligodendroglial myelination requires astrocyte-derived lipids.

PMID: 28549068 | Indexed in PubMed

In the vertebrate nervous system, myelination of axons for rapid impulse propagation requires the synthesis of large amounts of lipids and proteins by oligodendrocytes and Schwann cells. Myelin membranes are thought to be cell-autonomously assembled by these axon-associated glial cells. Here, we report the surprising finding that in normal brain development, a substantial fraction of the lipids incorporated into central nervous system (CNS) myelin are contributed by astrocytes. The oligodendrocyte-specific inactivation of sterol regulatory element-binding protein (SREBP) cleavage-activating protein (SCAP), an essential coactivator of the transcription factor SREBP and thus of lipid biosynthesis, resulted in significantly retarded CNS myelination; however, myelin appeared normal at 3 months of age. Importantly, embryonic deletion of the same gene in astrocytes, or in astrocytes and oligodendrocytes, caused a persistent hypomyelination, as did deletion from astrocytes during postnatal development. Moreover, when astroglial lipid synthesis was inhibited, oligodendrocytes began incorporating circulating lipids into myelin membranes. Indeed, a lipid-enriched diet was sufficient to rescue hypomyelination in these conditional mouse mutants. We conclude that lipid synthesis by oligodendrocytes is heavily supplemented by astrocytes in vivo and that horizontal lipid flux is a major feature of normal brain development and myelination.
